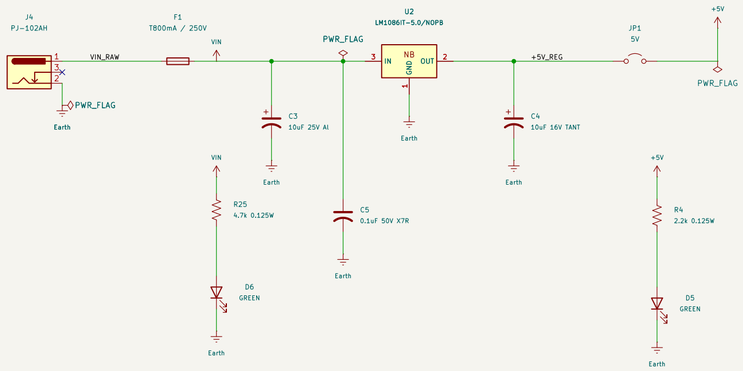
Fill-A-Bot Main Control Board

Power Input and 5V Regulator

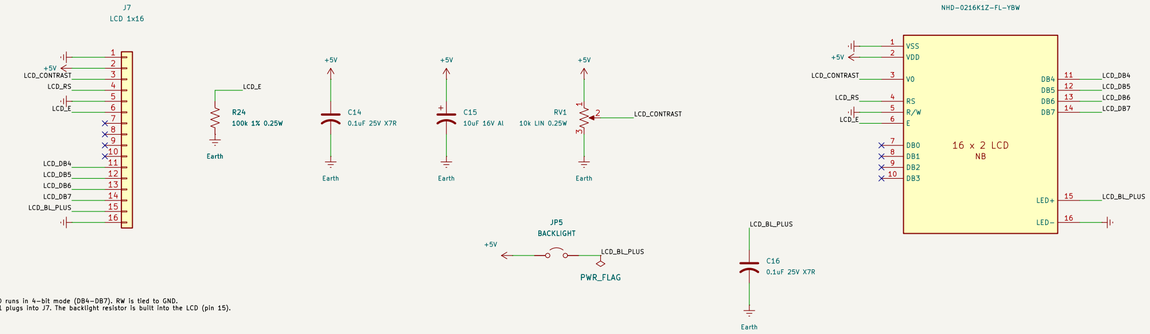


F1 is a 5x20mm fuse in a holder so it can be swapped.

Input is 9–12V DC, center positive.
U2 gets a heat sink (HSE14-38+22SP).
D5 and D6 light up when the 5V and 9V rails are on.

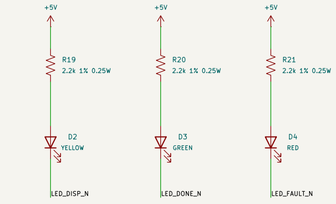
TP1 VIN_RAW · TP2 VIN · TP3 +5V_REG · TP4 +5V · TP5 Earth

16x2 LCD

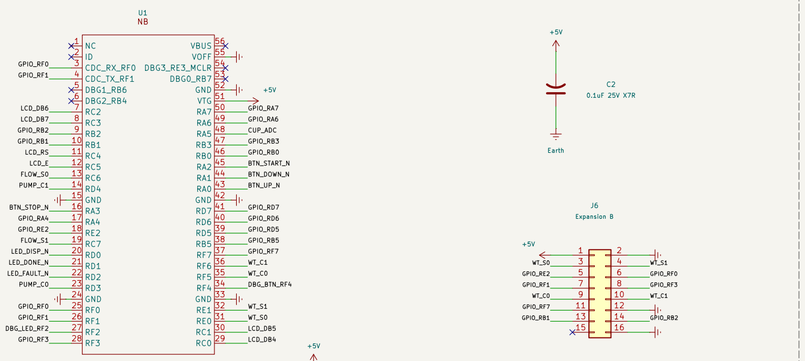


LCD runs in 4-bit mode (DB4–DB7), RW is tied to GND.
DS1 plugs into J7. The backlight resistor is built into the LCD (pin 15).

Status LEDs

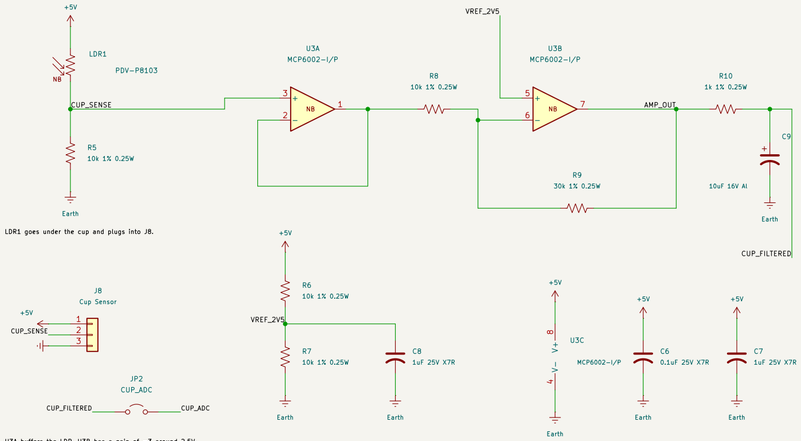


Curiosity Nano, Extra Headers, Debug



Extra headers for spare pins

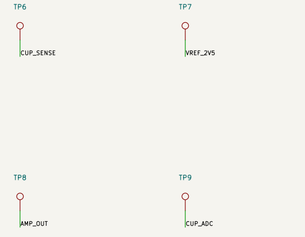
Cup Sensor (LDR + Op Amp)



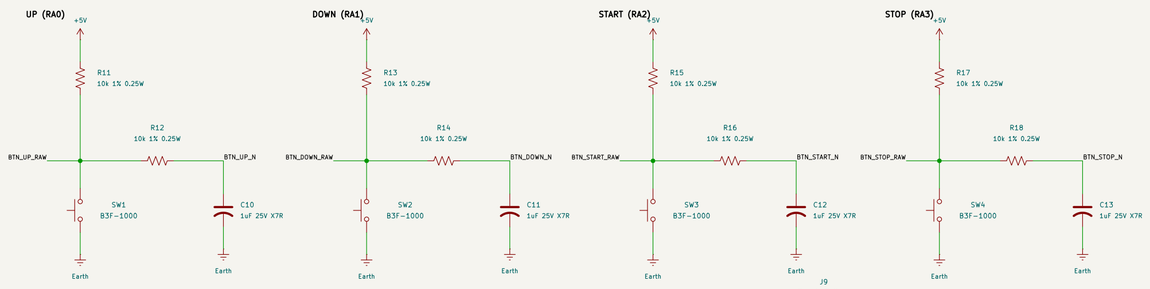
LDR1 goes under the cup and plugs into J8.

U3A buffers the LDR. U3B has a gain of −3 around 2.5V.
R10 and C9 make a low-pass filter (about 16 Hz).

Test Points



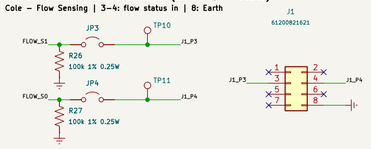
Buttons



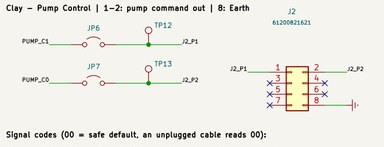
Each button has a 10k pull-up and a 10k/1uF RC debounce.
J9 is for buttons mounted off the board.

Ribbon Cables to Teammates (2-wire, 4-state)

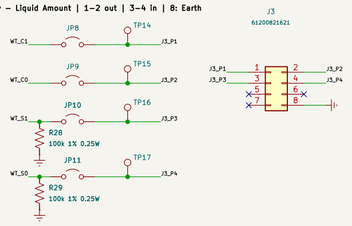
Cole — Flow Sensing | 3–4: flow status in | 8: Earth



Clay — Pump Control | 1–2: pump command out | 8: Earth



Troy — Liquid Amount | 1–2 out | 3–4 in | 8: Earth



Signal codes (00 = safe default, an unplugged cable reads 00):

J1 flow status: 00 no flow, 01 flowing, 10 no flow, 11 sensor fault

J2 pump command: 00 stop, 01 run, 10 and 11 stop

J3 weight command: 00, 01, 10, 11 = preset 1 to 4

J3 weight status: 00 not ready, 01 below target, 10 reached, 11 error

R26–R29: 100k pull-downs on my four ribbon inputs

Nikita Bangiyev — EGR 304 Team 201

Sheet: /
File: EGR304_Main_Control_Nikita.kicad_sch

Title: Fill-A-Bot Main Control Board

Size: A2	Date: 2026-10-30	Rev: 4
KiCad E.D.A. 10.0.7		Id: 1/1